

PENTICTON, BC, Canada

WMO: 718890

Lat: 49.460N

Lon: 119.600W

Elev: 344

StdP: 97.26

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.8	-10.0	-19.5	0.7	-11.8	-16.3	0.9	-8.3	12.8	3.6	11.4	2.7	3.4	350	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	33.0	18.6	31.2	18.1	29.5	17.4	19.6	30.4	18.7	29.0	17.9	27.9	4.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	11.6	23.0	14.7	10.9	22.5	13.7	10.2	22.2	57.4	30.4	54.4	29.1	51.9	28.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.2	8.2	DB	-15.6	36.4	2.8	1.1	-17.6	37.2	-19.2	37.8	-20.8	38.4	-22.9	39.2	(4)
(5)			WB	-16.3	21.2	2.7	1.0	-18.3	21.9	-19.8	22.5	-21.3	23.1	-23.3	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-0.5	0.6	4.9	9.1	14.2	17.6	21.6	20.8	15.6	8.9	3.4	-0.4
	DBStd	8.62	4.43	4.14	3.62	3.06	3.45	3.18	3.05	2.88	3.09	3.34	4.28	4.36
	HDD10.0	1387	324	262	161	52	6	0	0	0	1	59	199	323
	HDD18.3	3386	582	495	417	276	135	50	7	11	92	291	447	582
	CDD10.0	1284	0	0	3	26	135	229	361	334	169	26	2	0
	CDD18.3	241	0	0	0	0	6	29	110	86	10	0	0	0
	CDH23.3	3113	0	0	0	3	134	398	1374	1056	147	1	0	0
	CDH26.7	1156	0	0	0	0	30	123	578	394	31	0	0	0
Wind	WSAvg	3.2	4.2	3.5	3.3	2.9	2.7	2.6	2.7	2.6	2.6	3.1	4.2	4.4
Precipitation	PrecAvg	352	32	20	22	24	39	48	25	22	20	31	32	37
	PrecMax	483	68	35	51	64	88	95	84	77	71	86	60	61
	PrecMin	221	15	6	7	3	6	11	0	1	1	7	8	16
	PrecStd	64	14	9	12	16	20	21	18	20	18	18	14	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	10.8	16.7	23.1	29.7	33.1	35.8	34.6	30.0	20.8	14.2	9.7
		MCWB	4.8	6.2	9.1	12.7	16.7	18.5	19.6	18.8	17.5	13.5	10.4	6.5
	2%	DB	6.6	8.8	14.4	20.1	26.9	30.0	33.8	32.3	27.0	18.2	12.1	7.5
		MCWB	4.2	5.1	7.9	11.1	15.3	17.2	18.8	18.4	16.5	12.0	8.8	4.8
	5%	DB	5.4	7.1	12.6	17.8	24.3	27.6	32.0	30.6	24.8	16.5	10.6	6.0
		MCWB	3.5	4.1	7.0	9.8	14.0	16.3	18.4	17.9	15.5	10.9	7.8	3.9
	10%	DB	4.4	5.9	10.9	15.9	22.0	25.4	30.0	29.0	22.8	15.0	9.3	4.7
		MCWB	2.8	3.4	6.0	8.6	13.0	15.2	17.6	17.4	14.3	9.9	6.6	2.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	7.4	10.1	13.3	17.4	19.6	21.2	20.4	18.6	14.5	10.9	7.1
		MCDB	7.5	10.0	14.9	21.6	27.8	30.2	32.6	31.8	27.1	18.7	13.6	9.0
	2%	WB	4.5	5.6	8.6	11.7	15.9	18.0	20.0	19.3	17.2	13.0	9.3	5.3
		MCDB	6.2	7.8	13.3	18.8	24.9	28.3	30.9	29.9	25.0	17.4	11.5	7.1
	5%	WB	3.7	4.6	7.5	10.4	14.8	16.8	19.1	18.5	16.1	11.7	8.2	4.1
		MCDB	5.2	6.7	11.7	16.6	22.8	25.9	29.8	28.7	23.4	15.5	10.3	5.7
	10%	WB	2.9	3.7	6.5	9.2	13.6	15.8	18.2	17.7	15.0	10.5	6.9	3.0
		MCDB	4.3	5.6	10.4	14.9	20.8	24.1	28.4	27.7	22.0	14.2	9.0	4.5
Mean Daily Temperature Range	5% DB	MDBR	4.1	6.8	9.6	12.1	13.1	13.0	14.8	14.7	13.0	9.9	6.0	4.3
		MCDBR	4.8	7.8	12.2	15.3	17.6	17.0	18.0	17.5	16.3	11.9	7.1	5.3
		MCWBR	3.5	5.2	7.3	8.2	8.4	7.6	8.0	8.0	8.5	7.1	5.0	4.0
	5% WB	MCDBR	4.4	6.9	10.8	13.9	15.7	15.3	16.4	16.0	14.8	10.6	6.2	4.7
		MCWBR	3.3	4.6	6.8	7.9	7.7	7.1	7.5	7.6	7.9	6.6	4.5	3.7
Clear-Sky Solar Irradiance	taub	0.295	0.294	0.321	0.349	0.381	0.373	0.361	0.371	0.344	0.335	0.310	0.285	
	taud	2.311	2.367	2.392	2.385	2.335	2.388	2.445	2.420	2.482	2.481	2.443	2.380	
	Ebn at Noon	749	856	891	898	878	886	892	865	858	800	735	716	
	Edh at Noon	69	83	97	109	120	115	107	105	89	75	62	57	
All-Sky Solar Radiation	RadAvg	1.02	1.96	3.11	4.58	5.45	5.74	6.63	5.64	4.01	2.29	1.08	0.78	
	RadStd	0.12	0.20	0.30	0.35	0.43	0.51	0.52	0.41	0.35	0.28	0.13	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page